

INTERNATIONAL STANDARD

IEC 60297-5-107

First edition
2001-01

**Mechanical structures for electronic equipment –
Dimensions of mechanical structures
of the 482,6 mm (19 in) series –**

**Part 5-107:
Subracks and associated plug-in units –
Rear-mounted plug-in units**

*Structures mécaniques pour équipement électronique –
Dimensions des structures mécaniques de la série
de 482,6 mm (19 in) –*

*Partie 5-107:
Bacs et blocs enfichables associés –
Blocs enfichables à montage arrière*



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INTERNATIONAL ELECTROTECHNICAL COMMISSION

MECHANICAL STRUCTURES FOR ELECTRONIC EQUIPMENT – DIMENSIONS OF MECHANICAL STRUCTURES OF THE 482,6 mm (19 in) SERIES –

Part 5-107: Subracks and associated plug-in units – Rear mounted plug-in units

FOREWORD

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International Standard IEC 60297-5-107 has been prepared by subcommittee 48D: Mechanical structures for electronic equipment, of IEC technical committee 48: Electromechanical components and mechanical structures for electronic equipment.

The text of this standard is based on the following documents:

FDIS	Report on voting
48D/244/FDIS	48D/253/RVD

Full information on the voting for the approval of this standard can be found in the report on voting indicated in the above table.

This publication has been drafted in accordance with the ISO/IEC Directives, Part 3.

IEC 60297-5 consists of the following parts under the general title: Mechanical structures for electronic equipment – Dimensions of mechanical structures of the 482,6 mm (19 in) series:

- Part 5-100. Subracks and associated plug-in units – Design overview
- Part 5-101. Subracks and associated plug-in units – Injector/extractor handle
- Part 5-102. Subracks and associated plug-in units – Electromagnetic shielding provision
- Part 5-103. Subracks and associated plug-in units – Electrostatic discharge protection

Part 5-104, Subracks and associated plug-in units – Keying

Part 5-105, Subracks and associated plug-in units – Alignment and/or earth pin

Part 5-107, Subracks and associated plug-in units – Rear mounted plug-in units

The committee has decided that the contents of this publication will remain unchanged until 2004. At this date, the publication will be

- reconfirmed;
- withdrawn;
- replaced by a revised edition, or
- amended.

A bilingual version of this publication may be issued at a later date.

INTRODUCTION

This part of IEC 60297 is based on IEC 60297-3 (1984), its Amendment 1 (1992), and IEC 60297-4 (1995). It contains detail dimensions which ensure dimensional interchangeability of subracks with front mounted plug-in units per IEC 60297-4 requiring additional rear mounted plug-in units as defined in this standard. Front and/or rear subrack mounted plug-in units are mechanically identical (not electrically), however, the rear subrack mounted plug-in unit is mounted in a mirror image position to the front subrack mounted plug-in unit. The component side 1 (printed board reference plane) of the front mounted plug-in unit and the component side 1 of the rear mounted plug-in unit (printed board reference plane) are thus "in line". The first front mounted plug-in unit refers to the first printed board position on the left side of the subrack, viewed from the front. The first rear mounted plug-in unit refers to the first rear mounted plug-in unit position on the right side of the subrack, viewed from the rear.

Like many of the IEC 60297-3 and IEC 60297-4 subracks and associated plug-in unit basic dimensions, this standard also refers to the IEC 60603-2 connector series for its basic dimensional relationship.

This standard applies only to the mechanical structures for electronic equipment practices according to the IEC 60297 series.

MECHANICAL STRUCTURES FOR ELECTRONIC EQUIPMENT – DIMENSIONS OF MECHANICAL STRUCTURES OF THE 482,6 mm (19 in) SERIES –

Part 5-107: Subracks and associated plug-in units – Rear mounted plug-in units

1 Scope and object

This part of IEC 60297 covers extended features of subrack rear mounted plug-in units added to (front mounted) plug-in units and subracks according to IEC 60297-3 and IEC 60297-4. By implementing this extended feature to the subrack and plug-in units, a new subrack and plug-in unit type (incompatible with IEC 60297-3 and IEC 60297-4) is created.

The purpose of this standard is to specify dimensions which will ensure dimensional interchangeability of subracks and associated plug-in units using the extended function of rear subrack mounted plug-in units added to IEC 60297-3 and IEC 60297-4.

For mechanical and climatic tests refer to IEC 61587-1.

For electromagnetic shielding performance tests refer to IEC/TS 61587-3.

2 Normative references

The following normative documents contain provisions which, through reference in this text, constitute provisions of this part of IEC 60297. For dated references, subsequent amendments to, or revisions of, any of these publications do not apply. However, parties to agreements based on this part of IEC 60297 are encouraged to investigate the possibility of applying the most recent editions of the normative documents indicated below. For undated references, the latest edition of the normative document referred to applies. Members of IEC and ISO maintain registers of currently valid International Standards.

IEC 60297-3, *Dimensions of mechanical structures of the 482,6 mm (19 in) series – Part 3: Subracks and associated plug-in units*

IEC 60297-4, *Mechanical structures for electronic equipment – Dimensions of mechanical structures of the 482,6 mm (19 in) series – Part 4: Subracks and associated plug-in units – Additional dimensions*¹

IEC 60297-5-100, *Mechanical structures for electronic equipment – Dimensions of mechanical structures of the 482,6 mm (19 in) series – Part 5-100: Subracks and associated plug-in units – Design overview*

IEC 60297-5-101, *Mechanical structures for electronic equipment – Dimensions of mechanical structures of the 482,6 mm (19 in) series – Part 5-101: Subracks and associated plug-in units – Injector/extractor handle*

IEC 60297-5-105, *Mechanical structures for electronic equipment – Dimensions of mechanical structures of the 482,6 mm (19 in) series – Part 5-105: Subracks and associated plug-in units – Alignment and/or earth pin*

IEC 60603-2, *Connectors for frequencies below 3 MHz for use with printed boards – Part 2: Detail specification for two-part connectors with assessed quality, for printed boards, for basic grid of 2,54 mm (0,1 in) with common mounting features*

¹ There is a consolidated edition 1.1 (1999) that includes IEC 60297-4 (1995) and its amendment 1 (1999).

IEC 60917-1, *Modular order for the development of mechanical structures for electronic equipment practices – Part 1: Generic standard*

IEC 61076-2, *Connectors for use in d.c., low-frequency analogue and digital high speed data applications – Part 2: Circular connectors with assessed quality – Sectional specification*

IEC 61076-4-101, *Connectors with assessed quality, for use in d.c. low-frequency analogue and in digital high speed data applications – Part 4: Printed board connectors – Section 101: Detail specification for two-part connector modules having a basic grid of 2,0 mm for printed boards and backplanes in accordance with IEC 917*

IEC 61076-4-113, *Detail specification for two-part connectors having 5 rows with a grid of 2,54 mm for printed boards and backplanes in bus applications²*

IEC 61587-1, *Mechanical structures for electronic equipment – Tests for IEC 60917 and IEC 60297 – Part 1: Climatic, mechanical tests and safety aspects for cabinets, racks, subracks and chassis*

IEC/TS 61587-3, *Mechanical structures for electronic equipment – Tests for IEC 60917 and IEC 60297 – Part 3: Electromagnetic shielding performance tests for cabinets, racks and subracks*

3 Definitions

For the purpose of this part of IEC 60297, the definitions of IEC 60917-1 apply.

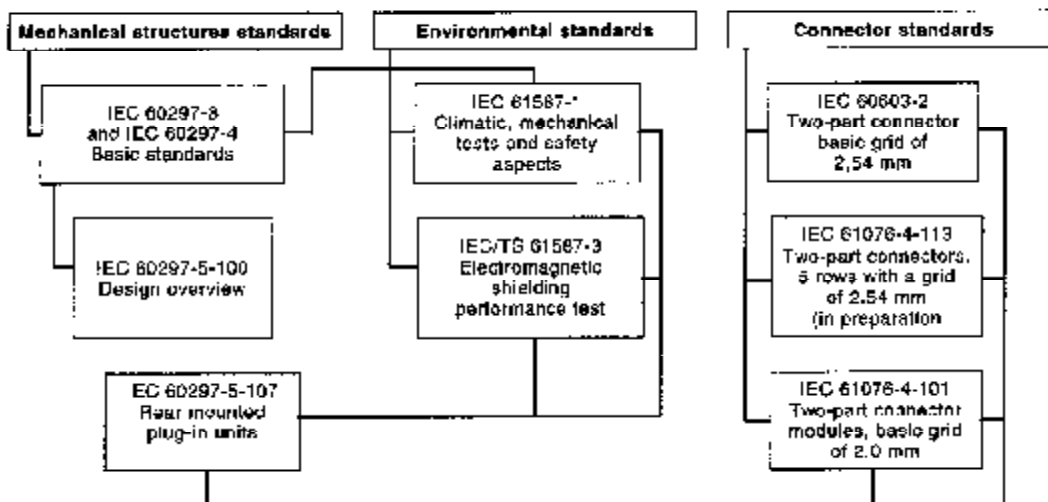
4 Extended feature added to IEC 60297-3 and IEC 60297-4 as described in IEC 60297-5-100

This standard gives dimensions only where they differ from or supplement those to be found in IEC 60297-3 and 60297-4. The dimensions used in this standard shall take precedence over those of IEC 60297-3 and 60297-4 when conformance to this standard is claimed. Dimensions shown in brackets are for reference only and are found in the stated standards.

The drawings in this standard are not intended to indicate product design.

Extended feature	Basic standards	Extended standards	Environmental standard(s)
Rear mounted plug-in units	IEC 60297-3 IEC 60297-4	IEC 60297-5-107	IEC 61587-1 IEC/TS 61587-3

² To be published.



5 General equipment arrangement

Generally, these are subracks featuring front and/or rear subrack mounted plug-in units.

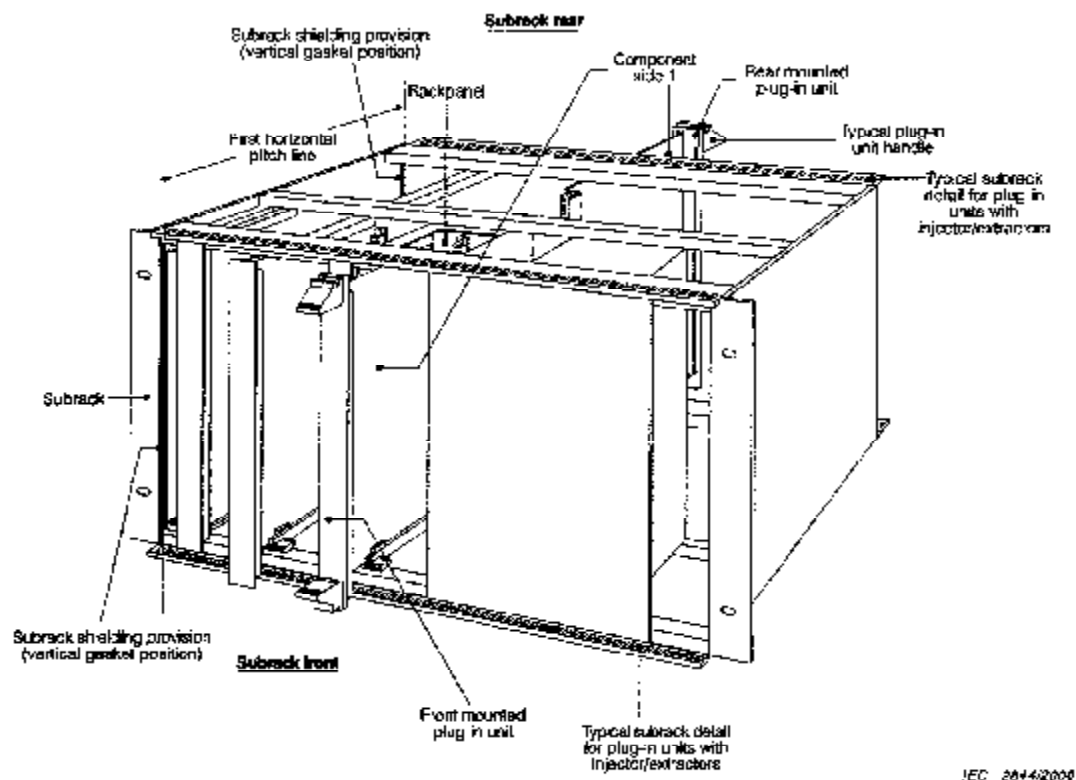


Figure 1 – General equipment arrangement – Typical 6U subrack with front and rear mounted plug-in units

6 Detail equipment arrangement

The detail arrangement below points out the added keying feature: the rear mounting of plug-in units in a subrack.

The RD_c dimension is measured from the rear attachment plane to the fixed connector (backplane) attachment plane.

The RD_x dimension is measured from the rear attachment plane to the bottom of the rear fixed connector shroud.

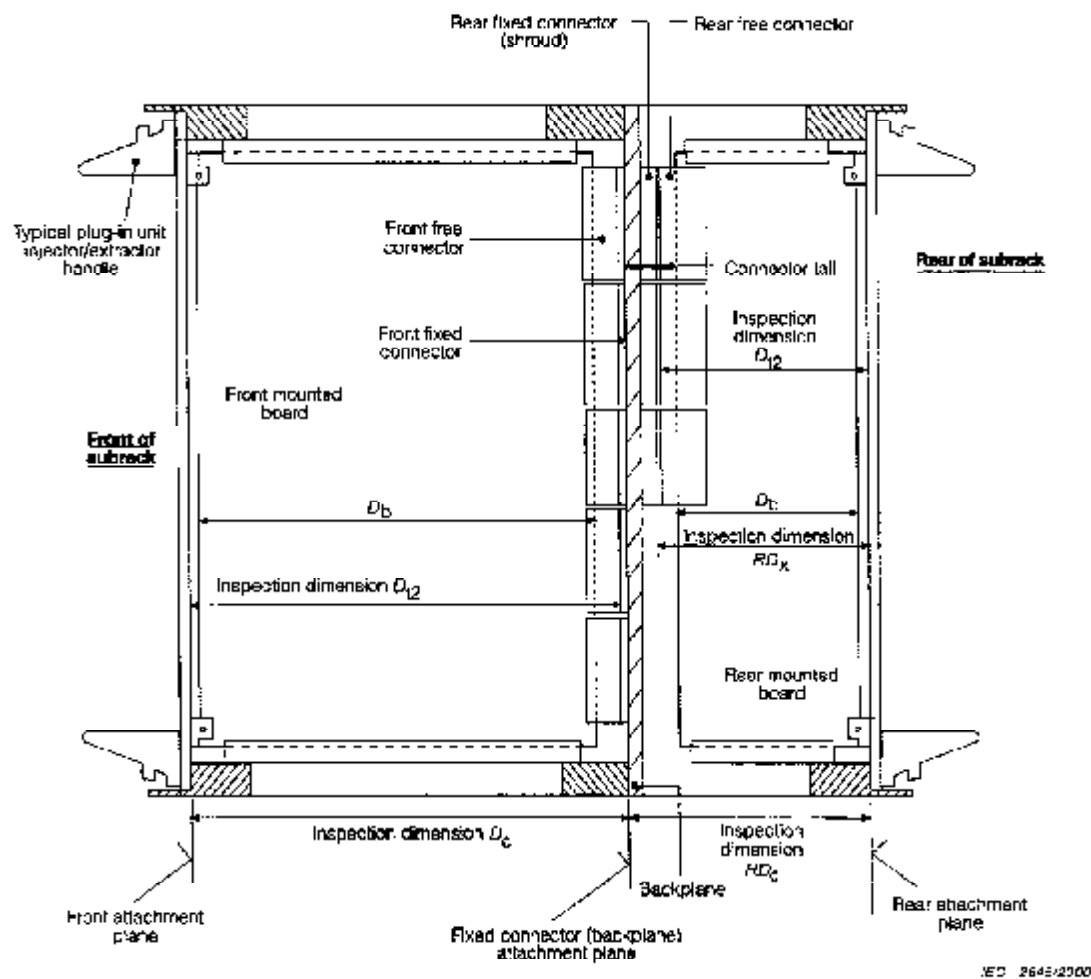


Figure 2 – Typical design detail equipment arrangement – Sectional side view (see figure 8)

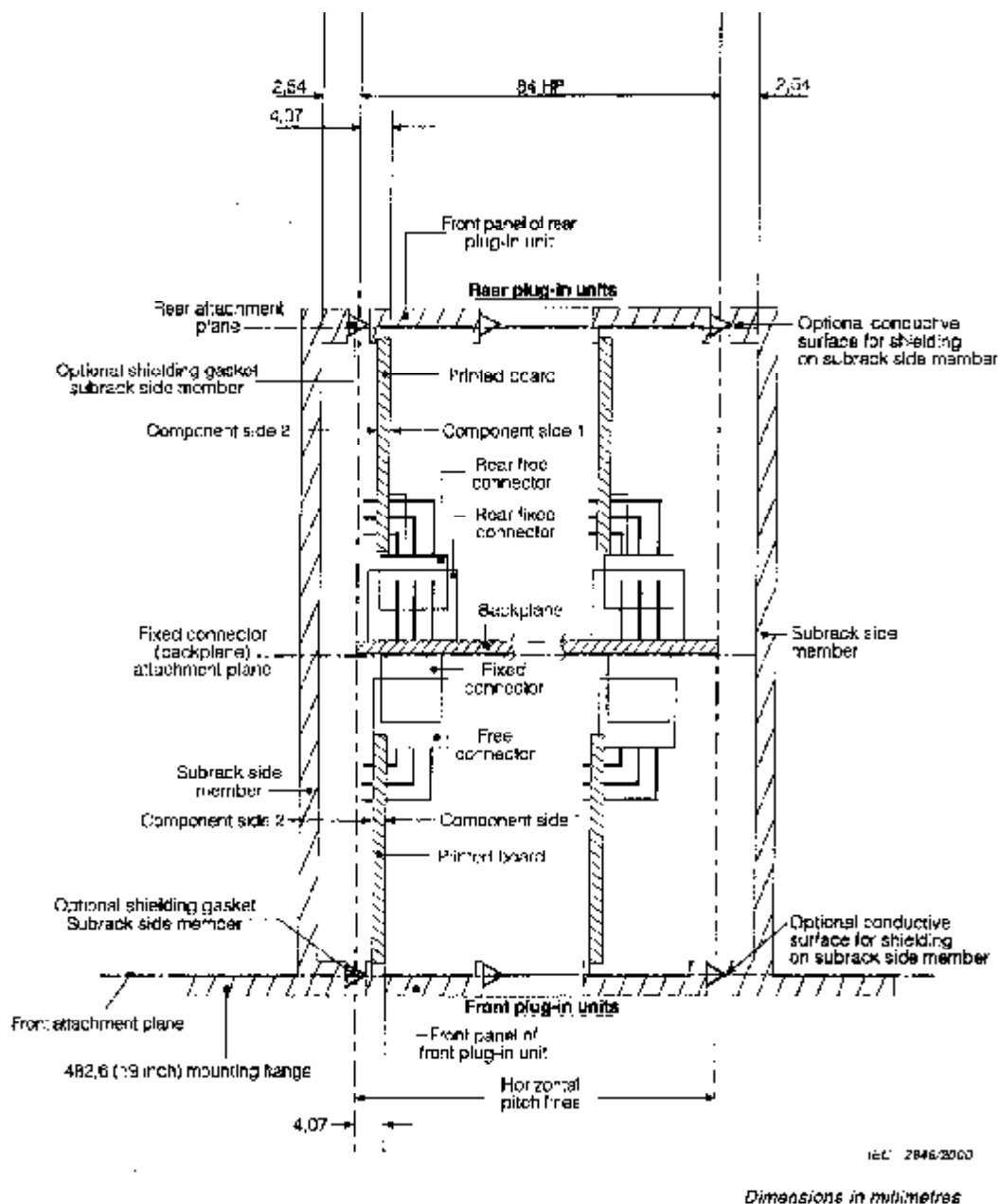


Figure 3 – Design detail equipment arrangement –
Sectional top view of a subrack assembled with plug-in units

7 Subrack rear mounted plug-in units

Rear mounted plug-in units are used, for example, for the management of rear subrack input/output (I/O) connections.

This standard defines a range of rear mounted plug-in unit printed board sizes and a subrack rear attachment plane for plug-in unit front panels.

According to IEC 60297-3 and IEC 60297-4, the subrack attachment plane/fixed connector attachment plane inspection dimension is D_c . This standard has specified an additional rear "attachment plane" inspection dimension RD_c suitable for subrack rear mounted plug-in units.

The rear attachment plane inspection dimension is based on the IEC 60603-2 connector with 17 mm tails, the IEC 61076-4-113 connector with 17 mm tails and the IEC 61076-4-101 connector with 16 mm tails. Other connectors with similar tail dimensions may be used, or other connector tail dimensions may be specified in a separate standard, resulting in a different RD_c and RD_c D_c dimensional relationship.

For example, in case of the IEC 61076-4-101 connector with a tail length of 13 mm or 14,5 mm the D_c dimension of the rear plug-in unit and the RD_c dimension of the subrack may increase by 2,5 mm. The RD_c dimension of the subrack does not change.

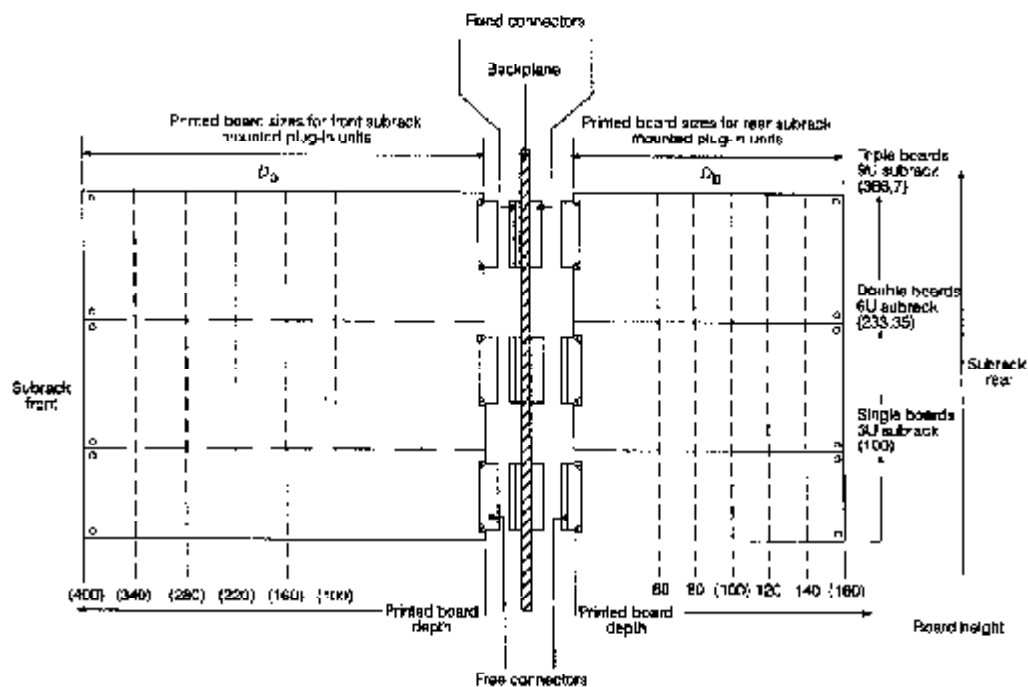
7.1 Rear mounted plug-in unit printed board dimensions

Table 1 – Additional rear mounted plug-in unit printed board dimensions

Dimensions in millimetres

$(D_b - 0,3)$	80	80	(100)	120	140	(160)
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1) See IEC 60297-3 and IEC 60297-4.



IEC 28472600

Dimensions in millimetres

Figure 4 – Subrack front/rear mounted plug-in unit printed board dimensions

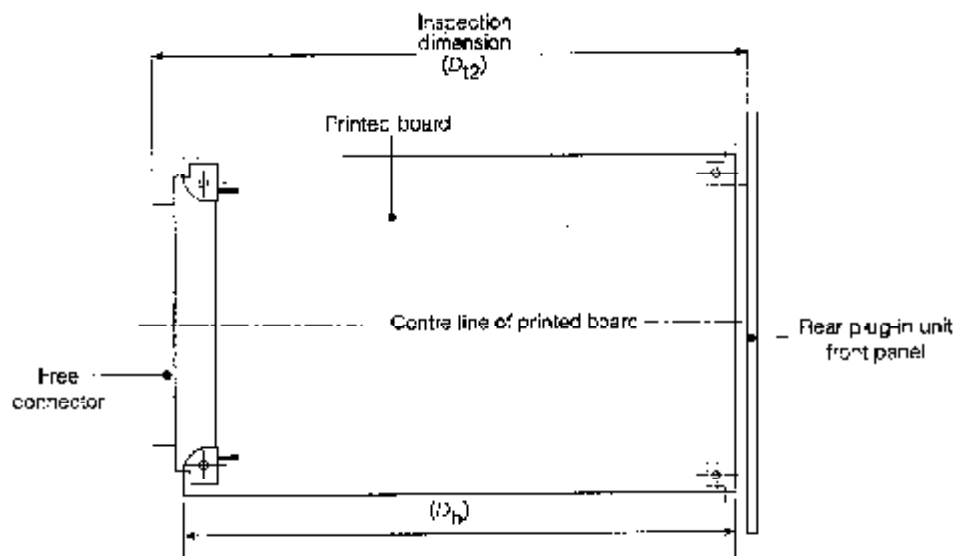
7.2 Subrack rear mounted plug-in units assembled with IEC 60603-2 and IEC 61076-4-113 connectors – Inspection dimensions

Table 2 – Additional rear mounted plug-in unit inspection dimensions assembled with IEC 60603-2 and IEC 61076-4-113 connectors

Dimensions in millimetres

(D_b)		80	90	(100)	120	140	(160)
$(D_{t2} = 0,4)$	IEC 60603-2	89,88	90,88	(109,88)	129,88	149,88	(169,88)
	IEC 61076-4-113	72,38	92,38	112,38	132,38	152,38	172,38

() See IEC 60297-3 and IEC 60297-4.



IEC 2846:2001.0

Figure 5 – Subrack rear mounted plug-in unit inspection dimensions for plug-in units assembled with IEC 60603-2 and IEC 61076-4-113 connectors

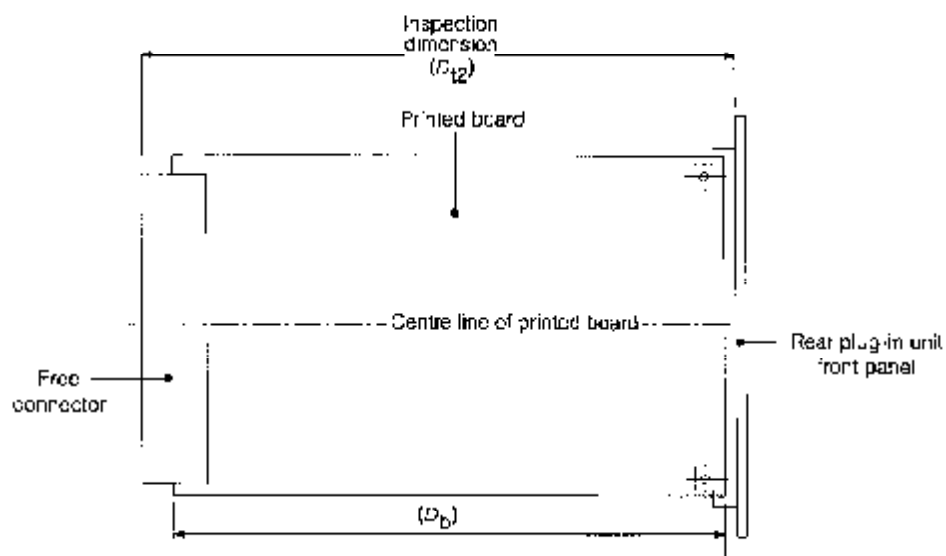
7.3 Subrack rear mounted plug-in units assembled with IEC 61076-4-101 connectors

Table 3 – Additional subrack front/rear mounted plug-in unit inspection dimensions assembled with IEC 61076-4-101 connectors

Dimensions in millimetres

(D_b)	60	80	(100)	120	140	(160)
$D_{t2} \pm 0,4$	71,74	91,74	111,74	131,74	151,74	171,74

() See IEC 60297-3 and IEC 60297-4.



IEC 3843:2000

Figure 5 – Subrack front/rear mounted plug-in unit inspection dimensions for plug-in units assembled with IEC 61076-4-101 connectors

7.4 Subrack dimensions for front/rear mounted plug-in units

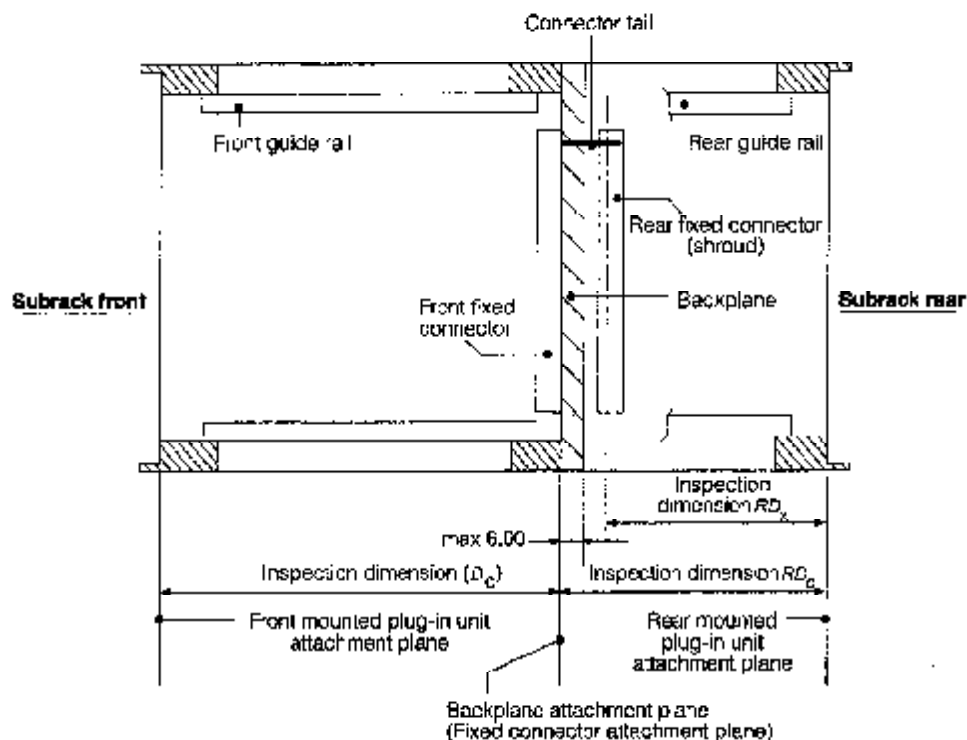
The inspection dimension RD_s is important for the user to inspect interchangeability with the appropriate plug-in unit. The inspection dimension RD_c is important for the designer of the subrack and back plane.

**Table 4 – Subrack inspection dimensions
for subrack rear mounted plug-in units (See figure 7 and figure 8)**

Dimensions in millimetres

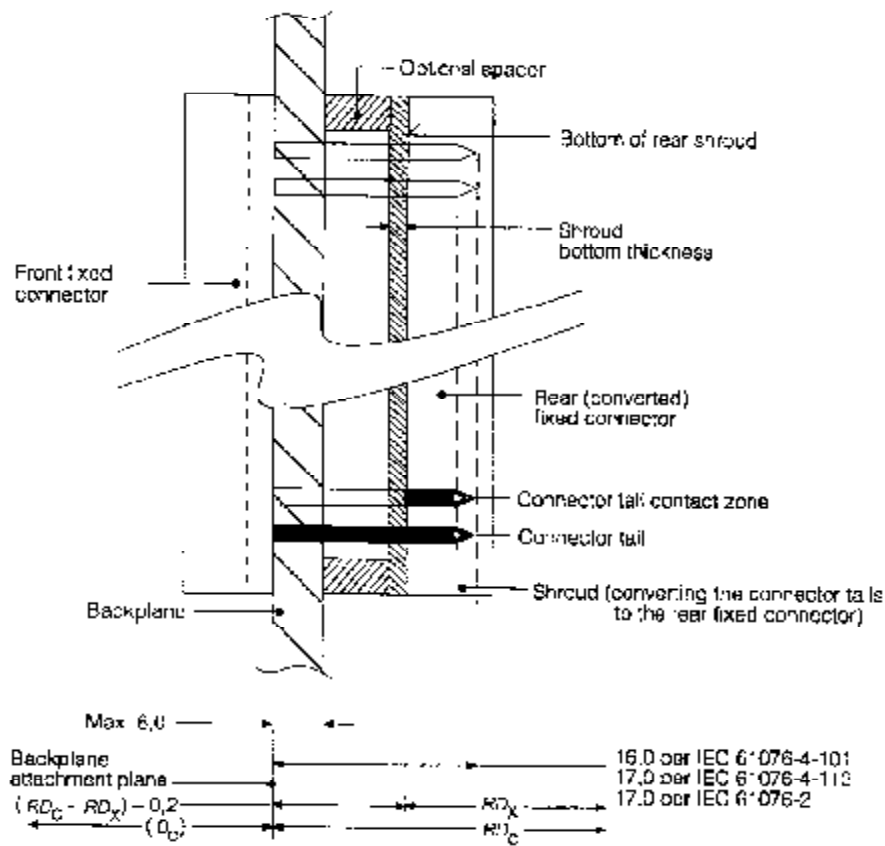
(D_s)		60	80	100	120	140	160
$RD_s \pm 0,5$		82,48	102,48	122,48	142,48	162,48	182,48
RD_s	IEC 60603-2	70,30	90,30	110,30	130,30	150,30	170,30
	IEC 61075-4-113	72,80	92,80	112,80	132,80	152,80	172,80
	IEC 61075-4-101	72,14	92,14	112,14	132,14	152,14	172,14

() See IEC 60297-3 and IEC 60297-4.



IEC 2852/2000

**Figure 7 – Subrack inspection dimensions for subrack rear mounted plug-in units
assembled with IEC 60603-2, IEC 61075-4-113 and IEC 61075-4-101 connectors**



IEC 2851:2000

Dimensions in millimetres

Figure 8 – Detail view of rear (converted) fixed connector

7.5 Rear subrack mounted plug-in unit guide rail orientation

Subrack rear mounted plug-in units are at mirror image to the front subrack mounted plug-in units. Since the first horizontal pitch (HP) line is found on the front mounted plug-in unit on the left side of the subrack (see IEC 60297-3 and IEC 60297-4), the first horizontal pitch line is to be found on the right side of the subrack when mounting plug-in units from the rear of the subrack.

() See IEC 60297-3 and IEC 60297-4.

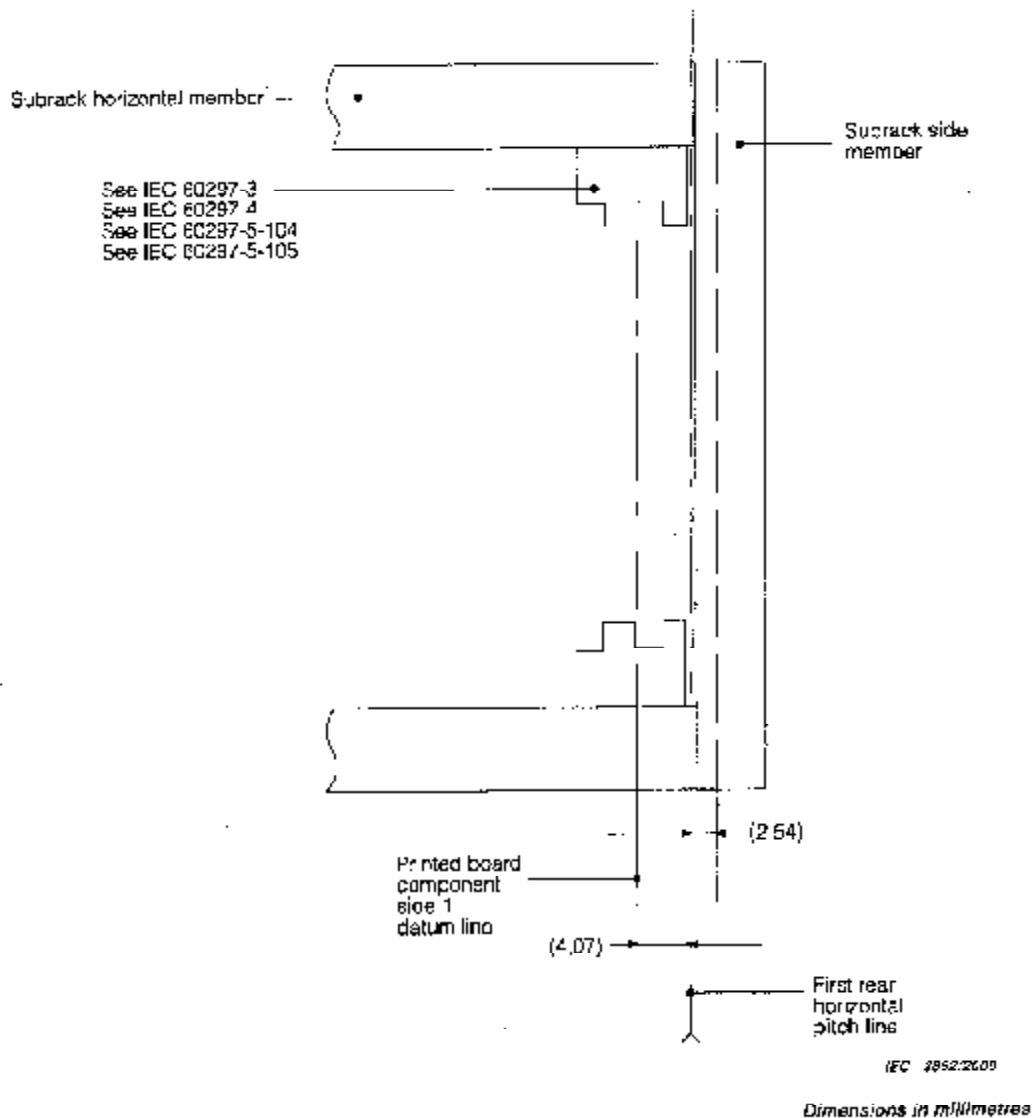


Figure 9 – Subrack rear mounted guide rail orientation



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